Math for CG

Linear Algebra

Outlines

- Matrix
- Vector
- Linear Algebra
- Linear Transformation

MATRIX

Column j

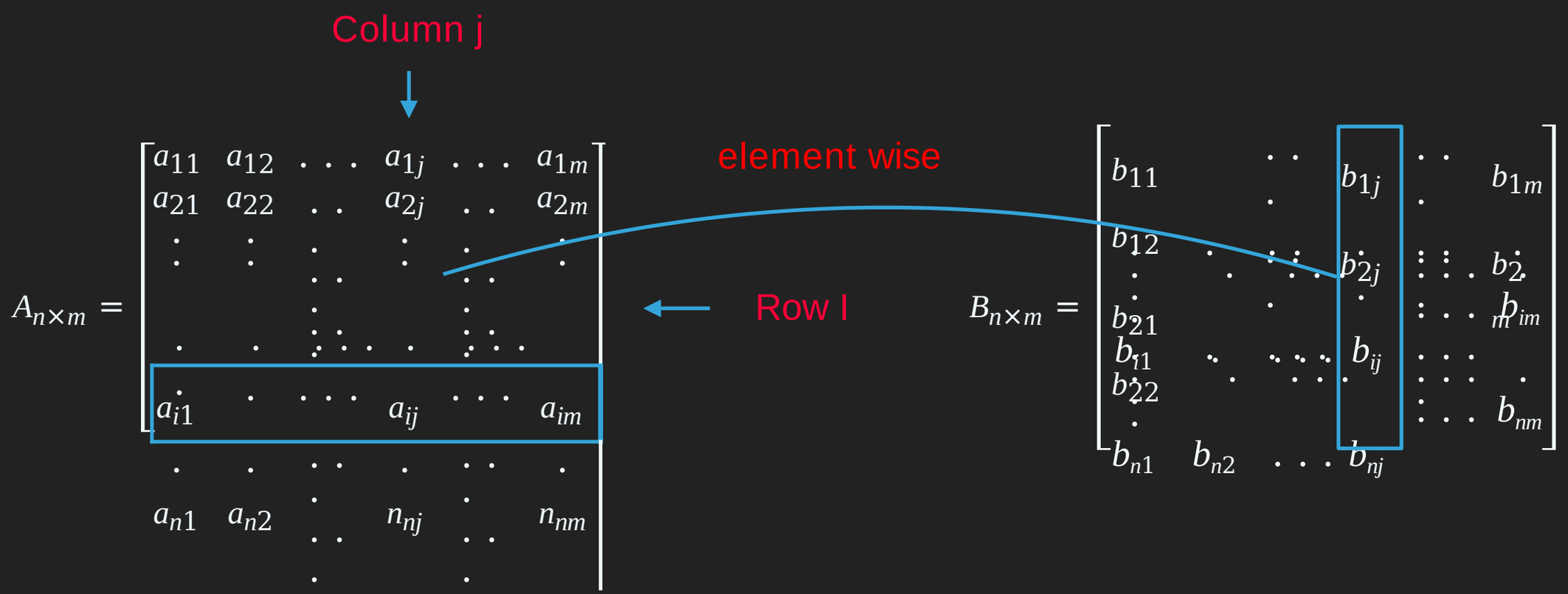
$$A_{n\times m} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1j} & \cdots & a_{1m} \\ a_{21} & a_{22} & \cdots & a_{2j} & \cdots & a_{2m} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ a_{i1} & a_{n2} & \cdots & a_{ij} & n_{nj} & a_{im} \\ a_{n1} & n_{nm} & & & & \end{bmatrix}$$

← Row i

$$\begin{bmatrix} 2 & 1 \\ 3 & 5 \end{bmatrix}$$

MATRIX ADDITION/SUBTRACTION

- Both matrix have the same shape (dimension)



$C_{n\times m} = A_{n\times m} +$

$B_{n\times m}$

$$\begin{bmatrix} 2 & 1 \\ 3 & 5 \end{bmatrix} + \begin{bmatrix} 6 & 2 \\ 1 & 5 \end{bmatrix} = \begin{bmatrix} 8 & 3 \\ 4 & 10 \end{bmatrix}$$

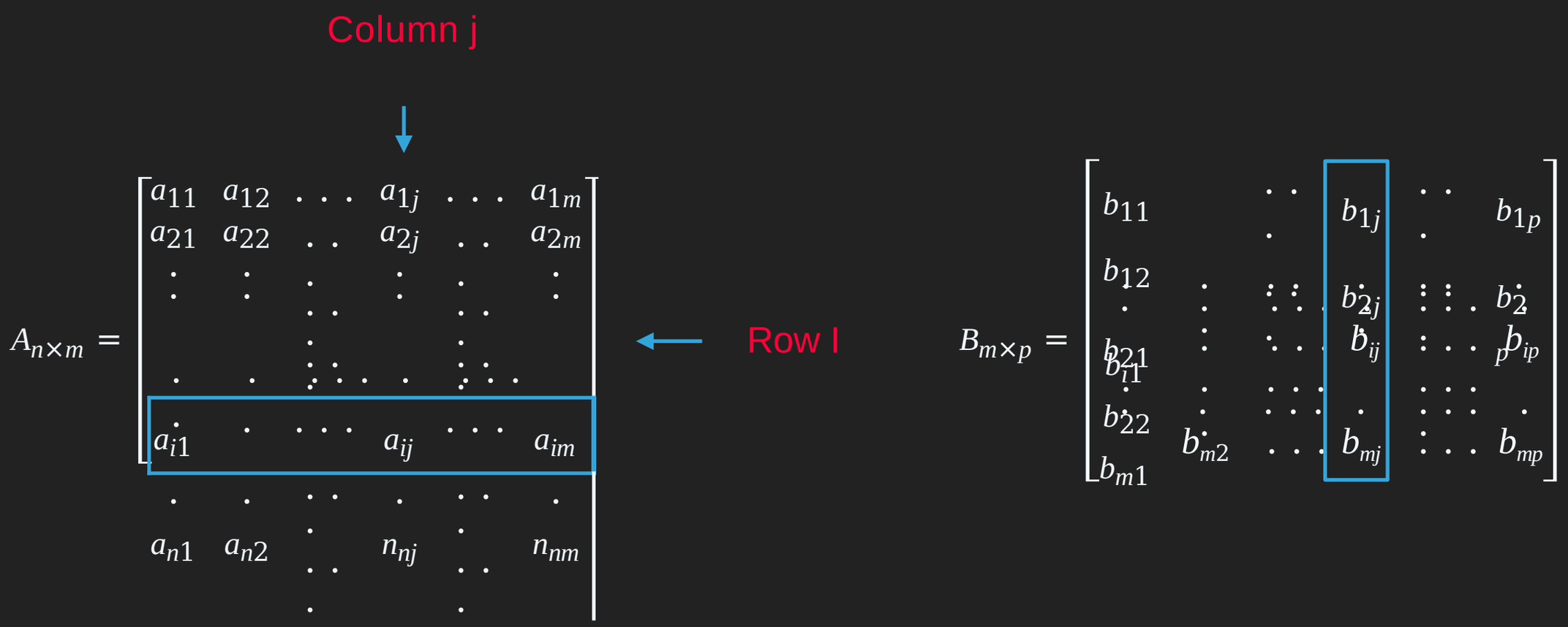
$C_{n\times m} = A_{n\times m} -$

$B_{n\times m}$

$$C_{ij} = a_{ij} - b_{ij}$$

MATRIX MULTIPLICATION

- Matrix



Matrix Multiplication: Product

$$C_{n\times p} = A_{n\times m} \\ B_{m\times p}$$

$$C_{ij} = \vec{a_{i.}} \cdot \vec{b_{.j}} = \sum_{k=1}^{k=m} a_{ik} \times b_{kj}$$

$$\begin{bmatrix} 2 & 1 \\ 3 & 5 \end{bmatrix} \begin{bmatrix} 6 & 2 \\ 1 & 5 \end{bmatrix} = \begin{bmatrix} 2\times 6 + 1\times 1 & 2\times 2 + 1\times 5 \\ 3\times 6 + 5\times 1 & 3\times 2 + 5\times 5 \end{bmatrix} = \begin{bmatrix} 13 & 9 \\ 23 & 29 \end{bmatrix}$$

Identity Matrix

$$I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}_{2\times2} \qquad I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}_{3\times3} \qquad I_4 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}_{4\times4}$$

$$A \times I = I \times A$$
$$= A$$

Inverse Matrix

$$I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}_{2\times2} \quad I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}_{3\times3} \quad I_4 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}_{4\times4}$$

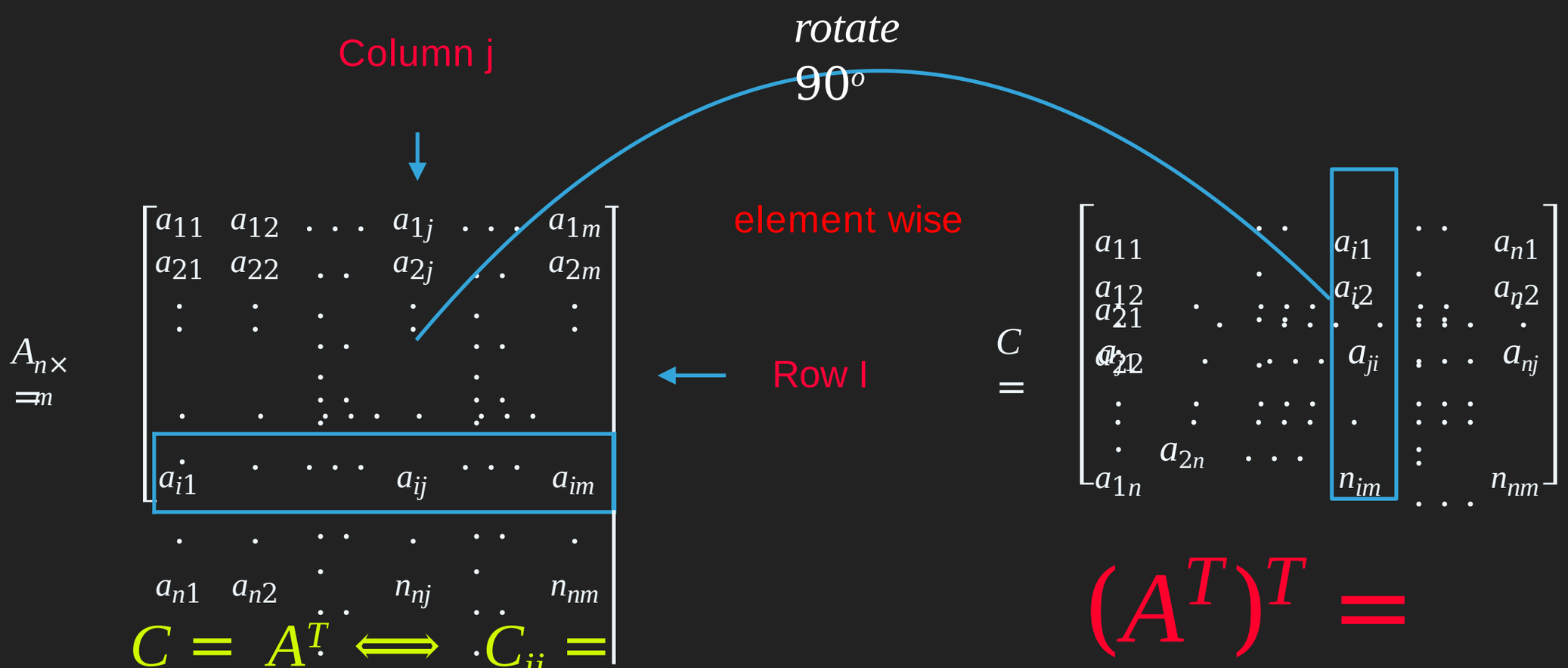
$AA^{-1} = A^{-1}A = I$

A must be square and $det(A) \neq 0$

$$A = \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix} \Longrightarrow A^{-1} = \begin{bmatrix} \cos(-\theta) & -\sin(-\theta) \\ \sin(-\theta) & \cos(-\theta) \end{bmatrix} = \begin{bmatrix} \cos(\theta) & \sin(\theta) \\ -\sin(\theta) & \cos(\theta) \end{bmatrix}$$

$$\because AA^{-1} = \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix} \begin{bmatrix} \cos(\theta) & \sin(\theta) \\ -\sin(\theta) & \cos(\theta) \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

TRANSPOSE



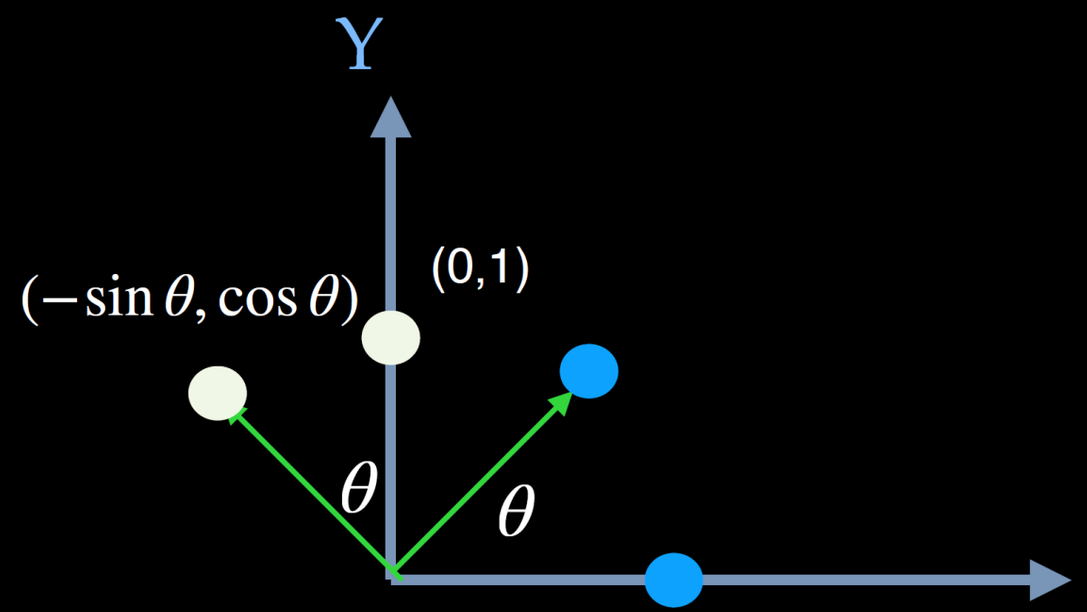
$$\begin{bmatrix} 2 & 1 \\ 3 & 5 \end{bmatrix}^T = \begin{bmatrix} 2 & 3 \\ 1 & 5 \end{bmatrix} \qquad \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}^T = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}$$

$$(A^T)^T =$$

$$(AB)^T = B^T A^T$$

Transformation

$$\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} \cos\theta \\ \sin\theta \end{bmatrix}$$



$$\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -\sin\theta \\ \cos\theta \end{bmatrix}$$

$$\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

Vectors

- A vector is a quantity with two
  - attributes Direction
  - Magnitude
- Examples include
  - Force
  - Velocity

$$Q = (x_2, y_2, z_2)$$

$$\boldsymbol{v}$$

$$P = (x_1, y_1, z_1)$$

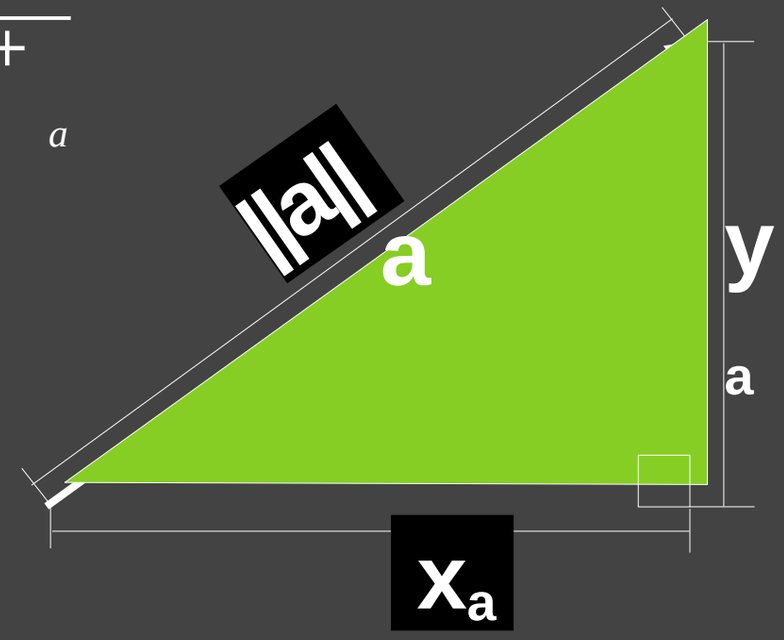
$$\overrightarrow{PQ} = Q - P = (x_2, y_2, z_2) - (x_1, y_1, z_1)$$

Vector Length

Vector a =(x_a, y_a)

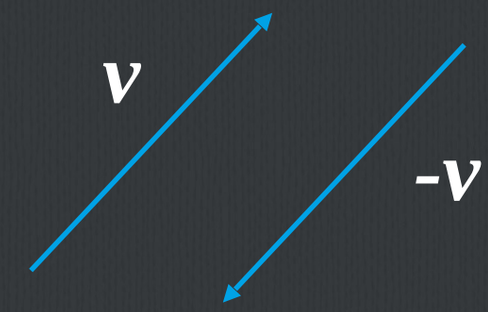
The length of vector a is computed as follow. $\|a\| = \sqrt{x_a^2 + y_a^2}$

$if v = (x, y, z)$ then $\|v\| = \sqrt{x^2 + y^2 + z^2}$

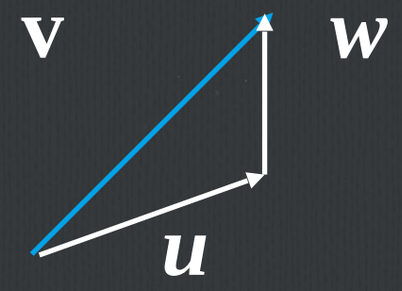


Vector Operations

- Every vector has an inverse
  - Same magnitude but points in opposite direction
- Every vector can be multiplied by a scalar







Vector Operations

$$V + W = (x_1, y_1) + (x_2, y_2) = (x_1 + x_2, y_1 + y_2)$$

$$V - W = (x_1, y_1) - (x_2, y_2) = (x_1 - x_2, y_1 - y_2)$$

$$\alpha V = \alpha(x, y) = (\alpha x, \alpha y)$$

Vector Transposition

A $vector$ $\vec{v} = (x_a, y_a, z_a)$ **express as matrix form**

$z_a)$

Given a vector(cc

$$v = \begin{bmatrix} x_a \\ y_a \\ z_a \end{bmatrix}$$

The transpose of vector v

$$v^T = \quad y_a \quad z_a]$$

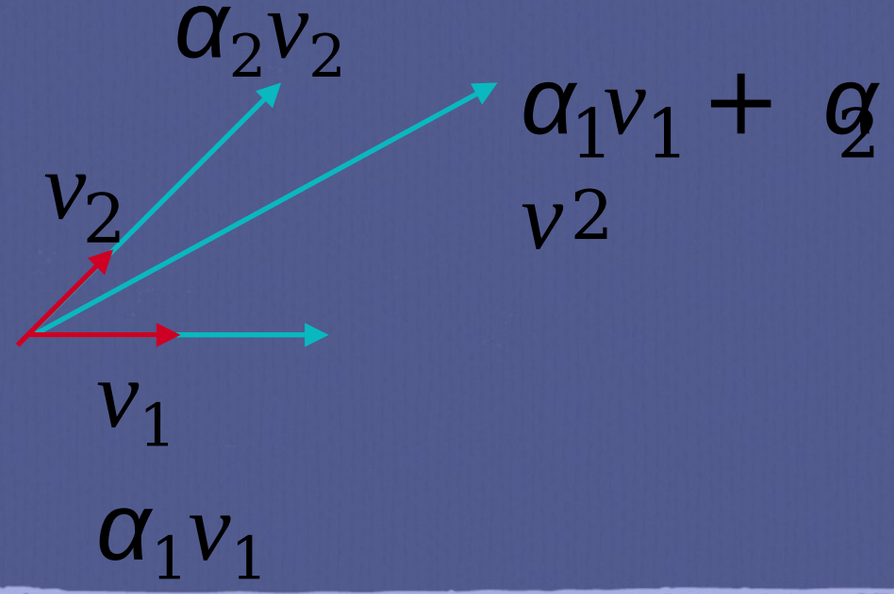
$$[x_a$$

Vector Space 向量空間

- 由一組線性獨立（linear independent）的向量稱為基底 $\{v_1, v_2, \ldots, v_n\}$ 所構成

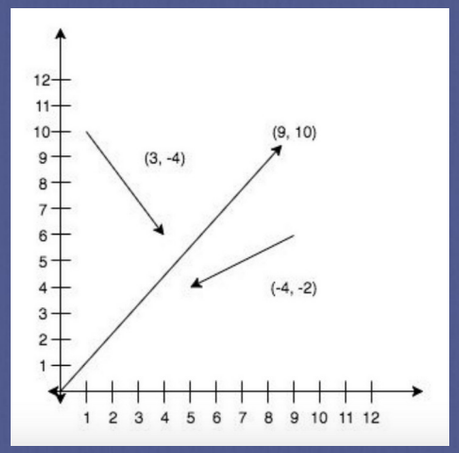
- 可經由向量加法 (vector addition) 與純量乘法 (Scale multiplication) 所擴展而成的線性空間

- $v = \alpha_1 v_1 + \alpha_2 v_2$



n-D Vector Space 向量空間

$$\vec{v} = c_1 v_1 + c_2 v_2 + \ldots + c_n v_n = [c_1, c_2, \ldots, c_n] \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix}$$



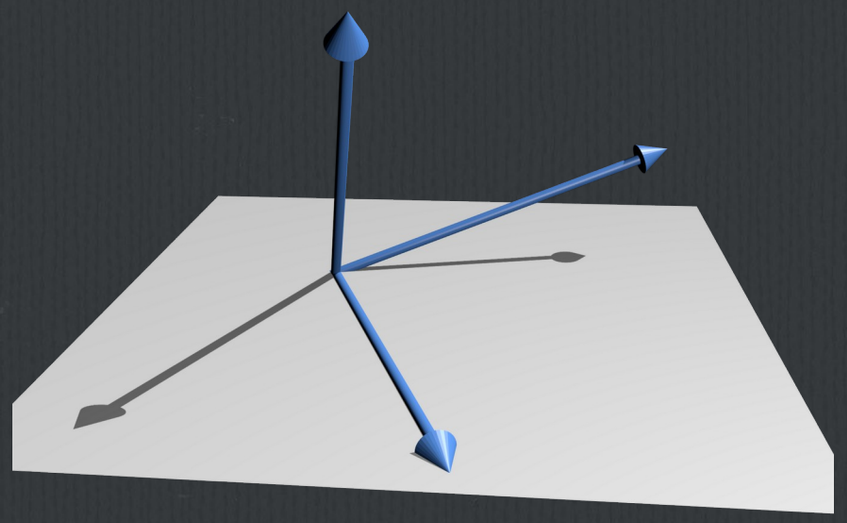
Linear Independence 線性獨立

- A set of vectors $v_1, v_2, \ldots, v_n$ is *linearly independent* if

$\rightarrow_1 v_1 + \rightarrow_2 v_2 + .. \rightarrow_n v_n = 0$ iff

$\rightarrow_1 = \rightarrow_2 = \ldots = 0$

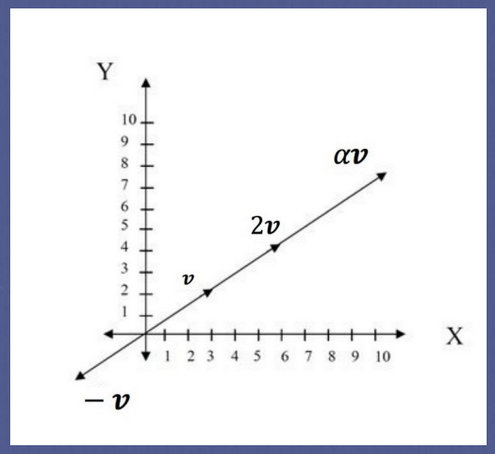


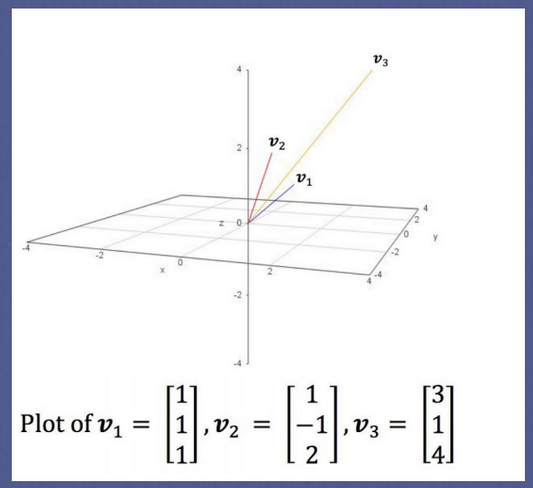


Linearly dependent vectors in a plane in $\mathbf{R}^3$.

Linear dependent

- If a set of vectors is linearly dependent, at least one can be written in terms of the others
  - Vector (3,6) and (1,2) are linear dependent
  - $v_1 = \begin{bmatrix}1\\1\\1\end{bmatrix}, v_2 = \begin{bmatrix}1\\-1\\2\end{bmatrix}, v_3 = \begin{bmatrix}3\\1\\4\end{bmatrix}$ are linear depenedent vectors





Linear independent

$$\overbrace{\underbrace{\begin{bmatrix}0\\0\\1\end{bmatrix}, \begin{bmatrix}0\\2\\-2\end{bmatrix}, \begin{bmatrix}1\\-2\\1\end{bmatrix}}_{\text{dependent}}, \begin{bmatrix}4\\2\\3\end{bmatrix}}^{\text{independent}}$$

$$\mathbf{v}_1 = \left(-\frac{5}{9}\right)\mathbf{v}_2 + \left(-\frac{4}{9}\right)\mathbf{v}_3 + \frac{1}{9}\mathbf{v}_4.$$

Dimension

- 在向量空間 (vector space) 中，線性獨立向量 (linearly independent vectors) 的最大數量是固定的，稱為空間維數 (Dimension)
- 在 n 維空間中， n 個線性獨立向量的任意集合構成該空間的基礎
- Given a basis $v_1, v_2, \ldots, v_n$, any vector v can be written as

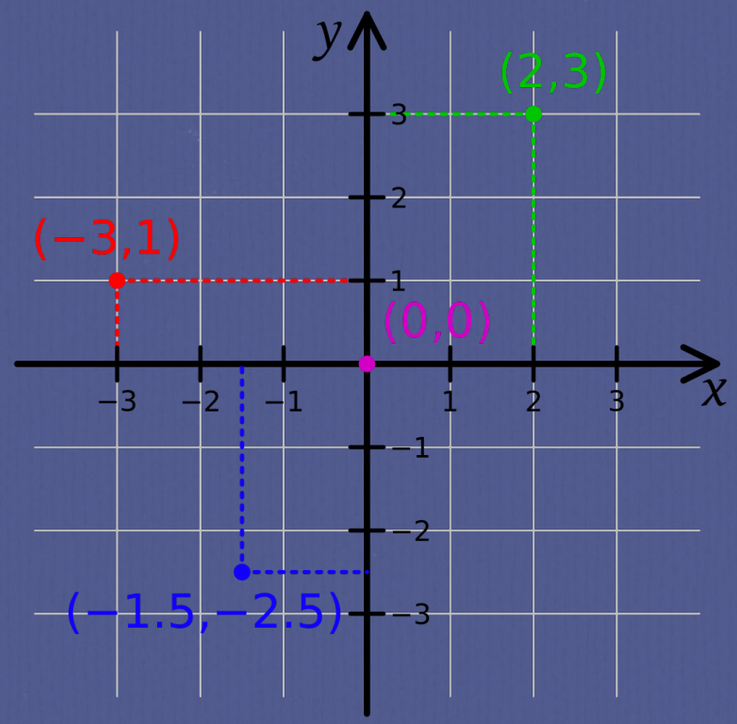
$$v = \rightarrow_1 v_1 + \rightarrow_2 v_2 + \ldots + \rightarrow_n v_n$$

$(\alpha_1, \alpha_2, \ldots, \alpha_n)$ is the presentation of v in this vector space

2D Vector Space 向量空間

The basis $\{e_1, e_2\}$ vector

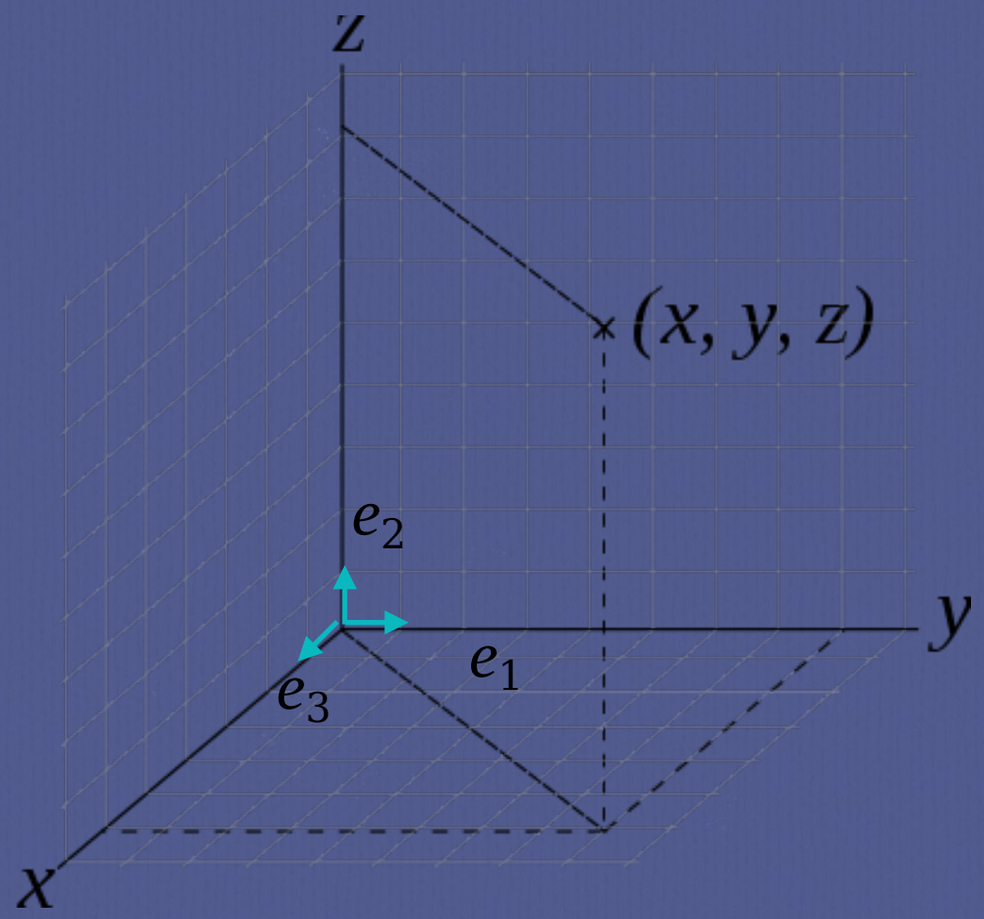
$(2,3) = 2(1,0) + 3(0,1)$



3D Vector Space 向量空間

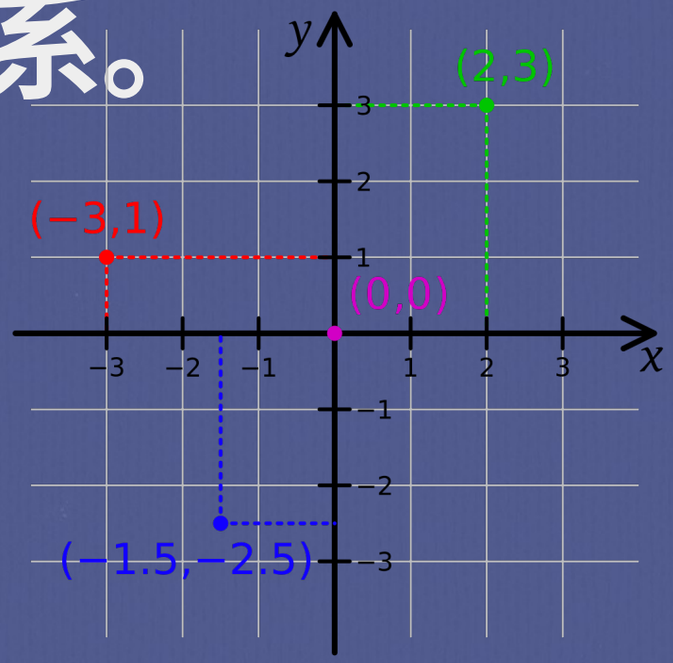
The basis $\{e_1, e_2, e_3\}$

vector $(x, y, z) = x(1,0,0) + y(0,1,0) + z(0,0,1)$



笛卡兒座標系（Cartesian coordinate system）

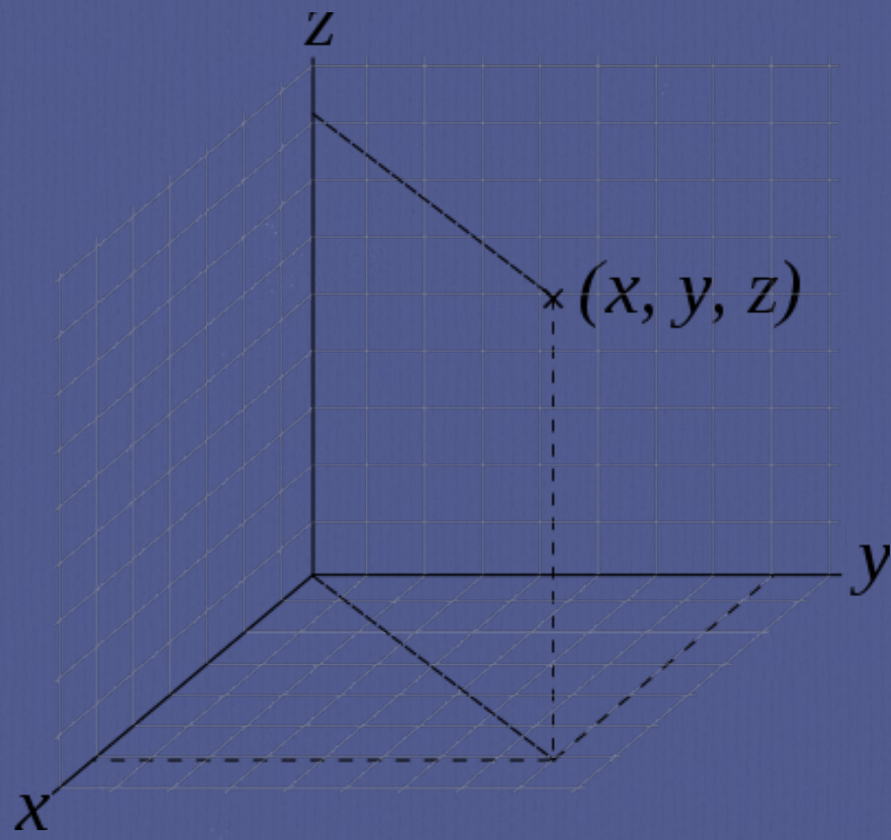
直角坐標系。



The basis $\{e_1, e_2\}$

The Origin O = (0,0)

vector (2,3) = 2(1,0) + 3(0,1)



The basis $\{e_1, e_2, e_3\}$

The Origin O=(0,0,0)

vector $(x, y, z) = x(1,0,0) + y(0,1,0) + z(0,0,1)$

2D 笛卡兒座標系（Cartesian coordinate system）

在平面上笛卡兒坐標為 (x_1, y_1) 和 (x_2, y_2) 的兩個點之間的歐幾里得距離是：

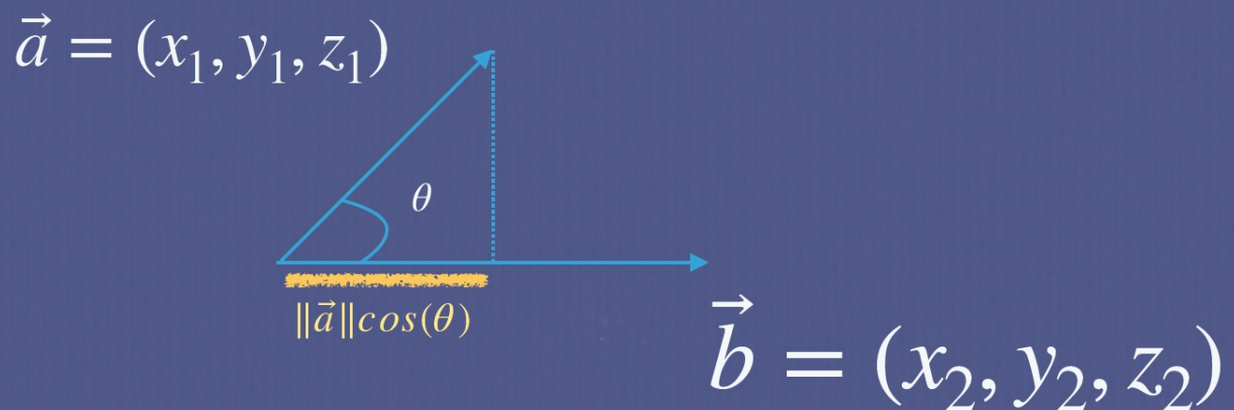
$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

在 3D 笛卡兒坐標為 (x_1, y_1, z_1) 和 (x_2, y_2, z_2) 的兩個點之間的歐幾里得距離是：

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

VECTOR INNER PRODUCT (DOT PRODUCT)

- $\vec{a} \cdot \vec{b} = \|\vec{a}\| \|\vec{b}\| cos(\theta) = (x_1, y_1, z_1) \cdot (x_2, y_2, z_2) = x_1 y_1 + x_2 y_2 + x_3 y_3$



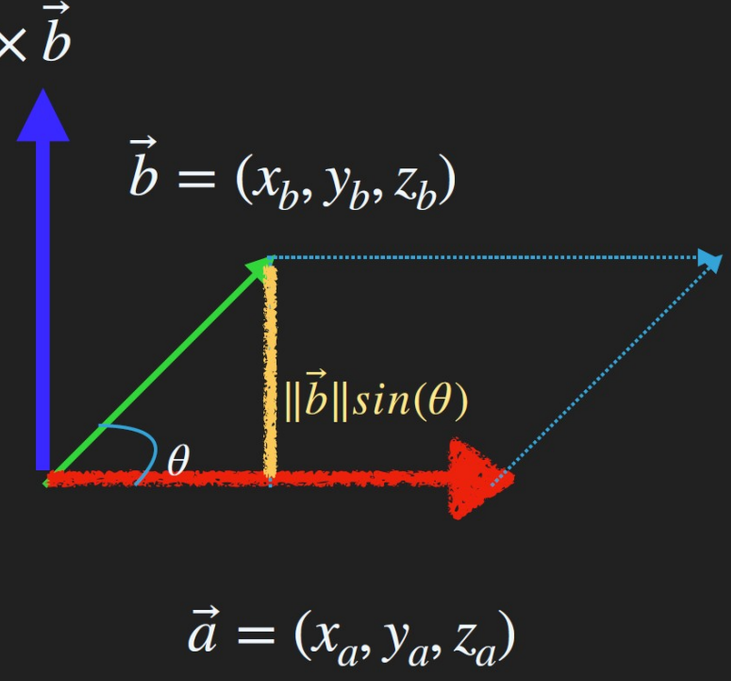
$\vec{a} \cdot \vec{b} = \|\vec{a}\| \|\vec{b}\| cos(\theta) = (x_1, y_1, z_1) \cdot (x_2, y_2, z_2) = x_1 y_1 + x_2 y_2 + x_3 y_3$

VECTOR CROSS PRODUCT

- $\vec{a} \times \vec{b} = (x_a, y_a, z_a) \times (x_b, y_b, z_b) = \left(\begin{vmatrix} y_a & z_a \\ y_b & z_b \end{vmatrix}, - \begin{vmatrix} x_a & z_a \\ x_b & z_b \end{vmatrix}, \begin{vmatrix} x_a & y_a \\ x_b & y_b \end{vmatrix} \right)$
- $\vec{a} \times \vec{b} = (x_a, y_a, z_a) \times (x_b, y_b, z_b) = (y_a z_b - y_b z_a \quad , x_b z_a - x_a z_b \quad , x_a y_b - x_b y_a)$
- $\vec{a} \times \vec{b} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ x_a & y_a & z_a \\ y_a & y_b & z_b \end{vmatrix}$

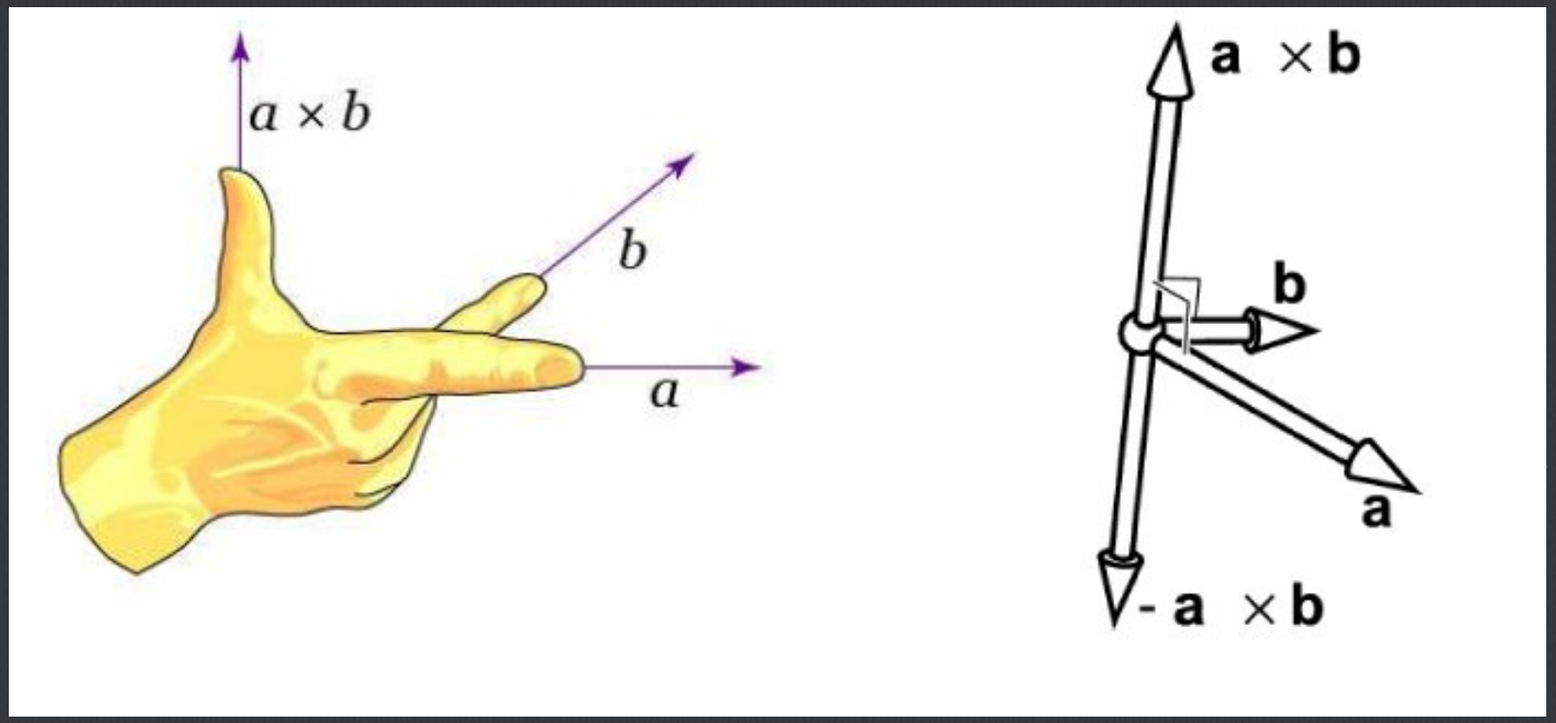
$\vec{a} \times \vec{b} \perp \vec{a}$

$\vec{a} \times \vec{b} \perp \vec{b}$



- $\|\vec{a} \times \vec{b}\| = \|\vec{a}\| \|\vec{b}\| sin(\theta) =$ 平行四行面積

Right Hand Rule



Examples

$$\vec{a} \times \vec{b} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ x_a & y_a & z_a \\ y_a & y_b & z_b \end{vmatrix}$$

$$\vec{a} \times \vec{b} = (x_a, y_a, z_a) \times (x_b, y_b, z_b)$$

$$= \left(\begin{vmatrix} y_a & z_a \\ b & z_b \end{vmatrix}, \begin{vmatrix} x_a & z_a \\ x_b & z_b \end{vmatrix}, \begin{vmatrix} x_a & y_a \\ x_b & y_b \end{vmatrix} \right)$$

$$= ($$

$$y$$

$$a = \begin{bmatrix} 3 \\ -3 \\ 1 \end{bmatrix} \quad b = \begin{bmatrix} 4 \\ 9 \\ 2 \end{bmatrix} \implies a \times b = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 3 & -3 & 1 \\ 4 & 9 & 2 \end{vmatrix} = \left(\begin{vmatrix} -3 & 1 \\ 9 & 2 \end{vmatrix}, - \begin{vmatrix} 3 & 1 \\ 4 & 2 \end{vmatrix}, \begin{vmatrix} 3 & -3 \\ 4 & 9 \end{vmatrix} \right) = (-15, -2{,}39)$$